STAT6611

- Lecture 11 (T, Sept 29)
- solutions posted to Homework 1
- Homework 2 assigned due F, Oct 2 by midnight (this week)

practice on LSE (without constraints)

- We'll have a 3rd homework before the midterm exam
- The midterm will be some type of take-home exam about 3 weeks from now (more details next time)

2.10 Extensions to Model III

First, we have the following lemma.

Lemma 2.15. *Corresponding to any $n \times n$ nonnegative definite matrix A of rank $r \geq 1$, there exists an $r \times n$ matrix S such that*

$$A = S'S.$$

Moreover, for any $r \times n$ matrix S such that $A = S'S$, $\text{rank}(S) = r$.

Proof. Since A is a symmetric matrix, there exists a matrix Q such that $Q'AQ = D$, where D is a diagonal matrix with the eigenvalues of A along the main diagonal, and where Q is an orthogonal matrix, that is, $Q'Q = QQ' = I$. (See, for example, Section 11.5 and Chapter 11A in Searle's book *Matrix Algebra Useful for Statistics*.) Because A is nonnegative definite, the diagonal entries of D must be non-negative and the number of positive entries must match the rank of A.

Since $\text{rank}(A) = r$, we may take $D = \begin{bmatrix} D_r & 0 \\ 0 & 0 \end{bmatrix}$, where D_r is an $r \times r$ nonsingular matrix, with all its diagonal elements being positive. Thus, we can define $D^{1/2}$ as the $r \times r$ nonsingular diagonal matrix whose diagonal elements are the square roots of those of D_r. Then,

$$A = QDQ' = Q \begin{bmatrix} D_r & 0 \\ 0 & 0 \end{bmatrix} Q' = Q \begin{bmatrix} D_r^{1/2} \\ 0 \end{bmatrix} \begin{bmatrix} D_r^{1/2} & 0 \end{bmatrix} Q' = S'S,$$

for an $r \times n$ matrix $S = \begin{bmatrix} D_r^{1/2} & 0 \end{bmatrix} Q'$.

Moreover, if S is any $r \times n$ matrix such that $A = S'S$, then by Lemma 2.3,

$$\text{rank}(S) = \text{rank}(S'S) = \text{rank}(A) = r.$$

□

In Model III, $\text{Var}(\underline{\epsilon}) = \sigma^2 H$, where H is a given nonnegative definite matrix.

Assume (first) that H is nonsingular to develop extensions to previous results.

H is invertible (n×n)

Write $H = R'R$ for an $n \times n$ nonsingular matrix R, where $H^{-1} = R^{-1}(R^{-1})'$.

Note that ← holds by Lem 2.15 + rank R = rank H = n

model III

$$\underline{y} = X\underline{\beta} + \underline{\epsilon} \iff (R^{-1})'\underline{y} = (R^{-1})'X\underline{\beta} + (R^{-1})'\underline{\epsilon}$$
$$\iff \underline{z} = W\underline{\beta} + \underline{h},$$

where

$$\underline{z} = (R^{-1})'\underline{y}, \qquad W = (R^{-1})'X, \qquad \underline{h} = (R^{-1})'\underline{\epsilon}$$

and

$$E[\underline{h}] = E[(R^{-1})'\underline{\epsilon}] = (R^{-1})'E(\underline{\epsilon}) = \underline{0}$$

with

$$\text{Var}[\underline{h}] = \text{Var}[(R^{-1})'\underline{\epsilon}] = (R^{-1})'(\sigma^2 H)R^{-1} = \sigma^2(R^{-1})'R'RR^{-1} = \sigma^2 I.$$

Var(ε)=Hσ², H=R'R

We will refer to the model

$$\underline{z} = W\underline{\beta} + \underline{h}$$

as the "transformed model," where the distribution of $\underline{h}$ is assumed to be identical to that of $(R^{-1})'\underline{\epsilon}$ under (the original) Model III and where the parameter space is taken to be $\{\underline{\beta}, \sigma^2 : \underline{\beta} \in \mathbb{R}^p, \sigma^2 > 0\}$.

The transformed model is of the same general form as Model II, with $E(\underline{h}) = \underline{0}$ and $\text{Var}(\underline{h}) = \sigma^2 I$.

Above one-to-one transformation between Model III $\underline{y} = X\underline{\beta} + \underline{\epsilon}$ & transformed (Model II) model $\underline{z} = W\underline{\beta} + \underline{h}$

Because $\underline{h}$ and $(R^{-1})'\underline{\epsilon}$ have the same distribution, the distribution of $\underline{z}$ or $\underline{y}$, as derived under the transformed model, is identical to the distribution of $\underline{z}$ or $\underline{y}$, as derived under Model III.

Note that $\mathcal{R}(W) = \mathcal{R}[(R^{-1})'X] = \mathcal{R}(X)$. Then, by Theorem 2.1,

a linear function $\underline{\lambda}'\underline{\beta}$ is estimable under the transformed model

$\iff$ $\underline{\lambda}' \in \mathcal{R}(W)$

$\iff$ $\underline{\lambda}' \in \mathcal{R}(X)$

$\iff$ $\underline{\lambda}'\underline{\beta}$ is estimable under Model III.

Furthermore, by Lemma 2.1, for any estimable function $\underline{\lambda}'\underline{\beta}$,

$c + \underline{a}'\underline{z}$ is a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$ under the transformed model

$\iff$ $c = 0$ and $\underline{\lambda}' = \underline{a}'W$ W is design matrix for z

$\iff$ $c = 0$ and $\underline{\lambda}' = \underline{a}'(R^{-1})'X$

$\iff$ $c + \underline{a}'(R^{-1})'\underline{y}$ is a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$ under Model III

$\iff$ $c + \underline{a}'\underline{z}$ is a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$ under Model III.

$$E(\underline{a}'\underline{z}) = \underline{a}'W\underline{\beta} = \underline{a}'(R^{-1})'X\underline{\beta} = E(\underline{a}'(R^{-1})'\underline{y})$$

Thus, the transformed model and Model III have the same set of estimable functions and the same set of linear unbiased estimators for every estimable $\underline{\lambda}'\underline{\beta}$.

Next we do LSE under transformed model

Take $\widehat{\underline{\beta}}$ to be any solution to the equations

NEs under transf. model

$$W'W\widehat{\underline{\beta}} = W'\underline{z}, \quad \text{or equivalently,} \quad X'H^{-1}X\widehat{\underline{\beta}} = X'H^{-1}\underline{y},$$

which are known as the Aitken equations. Here,

$$W'W = \left[(R^{-1})'X\right]'(R^{-1})'X = X'R^{-1}(R^{-1})'X = X'(R'R)^{-1}X = X'H^{-1}X$$

and

$$W'\underline{z} = \left[(R^{-1})'X\right]'(R^{-1})'\underline{y} = X'R^{-1}(R^{-1})'\underline{y} = X'(R'R)^{-1}\underline{y} = X'H^{-1}\underline{y}.$$

original NEs $X'X\underline{\beta} = X'\underline{y}$

If we substitute $H = I\sigma^2$ into Aitken equations, we get original NEs

By the Gauss-Markov theorem, under the transformed model, $\underline{\lambda}'\widehat{\underline{\beta}}$ is a linear unbiased estimator of an estimable function $\underline{\lambda}'\underline{\beta}$, and the variance of $\underline{\lambda}'\widehat{\underline{\beta}}$ is uniformly less than that of any other linear unbiased estimator. Thus, we have the following.

Theorem 2.19. *Under Model III, if $\underline{\lambda}'\underline{\beta}$ is estimable and $\widehat{\underline{\beta}}$ is any solution to the Aitken equations, then $\underline{\lambda}'\widehat{\underline{\beta}}$ is a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$, and the variance of $\underline{\lambda}'\widehat{\underline{\beta}}$ is uniformly less than that of any other linear unbiased estimator.*

that is, just transform model III (with general H) to model II - transformed with no loss of information & LSE is optimal with transformed data (original Gauss Markov theorem)

Note that $\widehat{\underline{\beta}}$ is a solution to the Aitken equations if and only if $(\underline{z}-W\underline{\beta})'(\underline{z}-W\underline{\beta})$ assumes a minimum at $\underline{\beta}=\widehat{\underline{\beta}}$, or equivalently if and only if $(\underline{y}-X\underline{\beta})'H^{-1}(\underline{y}-X\underline{\beta})$ assumes a minimum at $\underline{\beta}=\widehat{\underline{\beta}}$. The problem of minimizing

$$(\underline{y}-X\underline{\beta})'A(\underline{y}-X\underline{\beta})$$

with respect to $\underline{\beta}$ is called generalized least squares, where A is a positive definite matrix.

sum of squares for transformed data

If $\Lambda_1'\underline{\beta}$ and $\Lambda_2'\underline{\beta}$ are both estimable vectors and if $\widehat{\underline{\beta}}$ is a solution to the Aitken equations, then

$$\text{cov}(\Lambda_1'\widehat{\underline{\beta}},\Lambda_2'\widehat{\underline{\beta}})=\sigma^2\Lambda_1'(W'W)^-\Lambda_2=\sigma^2\Lambda_1'(X'H^{-1}X)^-\Lambda_2.$$

↑ previous variance under model II & $W=(R^{-1})'X$

Also, $p^*=\text{rank}(X)=\text{rank}(W)$. Thus, an unbiased estimator of σ^2 is

$$\widehat{\sigma}^2=(\underline{z}-W\widehat{\underline{\beta}})'(\underline{z}-W\widehat{\underline{\beta}})/(n-p^*)=(\underline{y}-X\widehat{\underline{\beta}})'H^{-1}(\underline{y}-X\widehat{\underline{\beta}})/(n-p^*).$$

↑ MSE from before based on transformed data

In Model III, $\mathrm{Var}(\underline{\epsilon}) = \sigma^2 H$, where H is a given nonnegative definite matrix.

Consider next that H is possibly singular. (H is not invertible)

For any linear estimator $c + \underline{a}'\underline{y}$ of an estimable function $\underline{\lambda}'\underline{\beta}$, we have

$$\mathrm{Var}(c + \underline{a}'\underline{y}) = \sigma^2 \underline{a}' H \underline{a}. \leftarrow \mathrm{Var}(\underline{a}'\underline{y})$$

Thus, the BLUE of an estimable function $\underline{\lambda}'\underline{\beta}$ is given by $\underline{a}'\underline{y}$ where $\underline{a}$ is any vector that minimizes $\underline{a}' H \underline{a}$ subject to the constraints $X'\underline{a} = \underline{\lambda}$.

↑ for $E(\underline{a}'\underline{y}) = \underline{\lambda}'\underline{\beta}$ need $X'\underline{a} = \underline{\lambda}$ or $\underline{\lambda}' = \underline{a}'X$

The Lagrangian function for this minimization problem is

weights $\underline{a}$ on $\underline{y}$ ↑ Lagrange multipliers

$$L(\underline{a}, \underline{\rho}) = \underline{a}' H \underline{a} + 2\underline{\rho}'(X'\underline{a} - \underline{\lambda}).$$

We have

$$\frac{\partial L}{\partial \underline{a}} = 2H\underline{a} + 2X\underline{\rho} \quad \text{and} \quad \frac{\partial L}{\partial \underline{\rho}} = 2(X'\underline{a} - \underline{\lambda}).$$

Equating these partial derivatives to zero gives

$$H\underline{a} + X\underline{\rho} = \underline{0} \quad \text{and} \quad X'\underline{a} = \underline{\lambda}.$$

Thus, a necessary condition for $\underline{a}' H \underline{a}$ to assume a minimum at $\underline{a} = \widetilde{\underline{a}}$, for $\underline{a}$ satisfying $X'\underline{a} = \underline{\lambda}$, is that the equations

$$\begin{bmatrix} H & X \\ X' & 0 \end{bmatrix} \begin{bmatrix} \underline{a} \\ \underline{\rho} \end{bmatrix} = \begin{bmatrix} \underline{0} \\ \underline{\lambda} \end{bmatrix} \qquad (1)$$

have a solution whose "first" component is $\widetilde{\underline{a}}$.

Take R to be any matrix such that $H = R'R$. [By Lemma 2.15, there exists an $r \times n$ matrix R such that $H = R'R$, where $r = \text{rank}(H)$.] Then, the above equations (1) can be written as

$$\begin{bmatrix} R'R & X \\ X' & 0 \end{bmatrix} \begin{bmatrix} \underline{a} \\ \underline{\rho} \end{bmatrix} = \begin{bmatrix} \underline{0} \\ \underline{\lambda} \end{bmatrix},$$

which look very much like the earlier constrained normal equations

$$\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} \begin{bmatrix} \widehat{\beta} \\ \underline{\rho} \end{bmatrix} = \begin{bmatrix} X'\underline{y} \\ \underline{b} \end{bmatrix}$$

upon replacing

X with R (and $X'X$ with $H = R'R$)

A' with X

$\widehat{\beta}$ with $\underline{a}$

$\underline{y}$ with $\underline{0}$

$\underline{b}$ with $\underline{\lambda}$

and $A\widehat{\beta} = \underline{b}$ with $X'\underline{a} = \underline{\lambda}$.

Thus, if $\underline{\lambda}'\underline{\beta}$ is estimable, then by Theorem 2.12, the equations (1) are consistent.

Theorem 2.20. *Under Model III, a linear estimator $c + \underline{a}'\underline{y}$ of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE (that is, has minimum variance in the class of linear unbiased estimators) if and only if $c = 0$ and $\underline{a}$ is the first component of some solution to the equations (1).*

Proof. Take $\begin{bmatrix} \underline{a} \\ \underline{\rho} \end{bmatrix}$ to be any solution of equations (1) so $H\underline{a} + X\underline{\rho} = \underline{0}$ and $X'\underline{a} = \underline{\lambda}$.

Thus, $\underline{\lambda}' = \underline{a}'X$, implying that $\underline{a}'\underline{y}$ is an unbiased estimator of $\underline{\lambda}'\underline{\beta}$ (by Lemma 2.1).

Take $k + \underline{b}'\underline{y}$ as any other linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$ so $k = 0$ and $X'\underline{b} = \underline{\lambda}$.

Note

$\text{cov}(\underline{b}'\underline{y}, \underline{a}'\underline{y}) = (\underline{b}-\underline{a})'\text{Var}(\underline{Y})\underline{a}$ ($\text{Var}(\underline{Y}) \sim \sigma^2 H$)

$= (\underline{b}-\underline{a})'\sigma^2 H\underline{a} = (-\sigma^2)(\underline{b}-\underline{a})'X\underline{\rho}$ ($H\underline{a} = -X\underline{\rho}$)

$= (-\sigma^2)(X'\underline{b} - X'\underline{a})'\underline{\rho} = 0$ ($X'\underline{b} = \underline{\lambda}$, $X'\underline{a} = \underline{\lambda}$)

Then, add/subtract $\underline{a}'\underline{y}$; $\text{cov}(\underline{b}'\underline{y}, \underline{a}'\underline{y}) = 0$

$$\text{Var}(k + \underline{b}'\underline{y}) = \text{Var}(\underline{b}'\underline{y}) = \text{Var}[\underline{a}'\underline{y} + (\underline{b} - \underline{a})'\underline{y}] = \text{Var}(\underline{a}'\underline{y}) + \text{Var}[(\underline{b} - \underline{a})'\underline{y}]$$

$$= \text{Var}(\underline{a}'\underline{y}) + (\underline{b} - \underline{a})'(\sigma^2 H)(\underline{b} - \underline{a}) = \text{Var}(\underline{a}'\underline{y}) + \sigma^2(\underline{b} - \underline{a})'R'R(\underline{b} - \underline{a}).$$

$\sigma^2(\underline{b} - \underline{a})'R'R(\underline{b} - \underline{a}) \geq 0$

Thus, $\text{Var}(k + \underline{b}'\underline{y}) \geq \text{Var}(c + \underline{a}'\underline{y})$, with equality holding iff $(\underline{b}-\underline{a})'R'R(\underline{b}-\underline{a}) = 0$

iff $R(\underline{b}-\underline{a}) = \underline{0} = R\underline{0}$

iff $R'R(\underline{b}-\underline{a}) = R'R\underline{0}$ (by Lemma 1.3)

iff $H(\underline{b}-\underline{a}) = \underline{0}$ or $H\underline{b} = H\underline{a}$ (since $H = R'R$)

iff $H\underline{b} + X\underline{\rho} = \underline{0}$ holds since $H\underline{a} + X\underline{\rho} = \underline{0}$ by assumption

iff $\begin{pmatrix} \underline{b} \\ \underline{\rho} \end{pmatrix}$ solves (1) too (since $X'\underline{b} = \underline{\lambda}$ by assumption)